

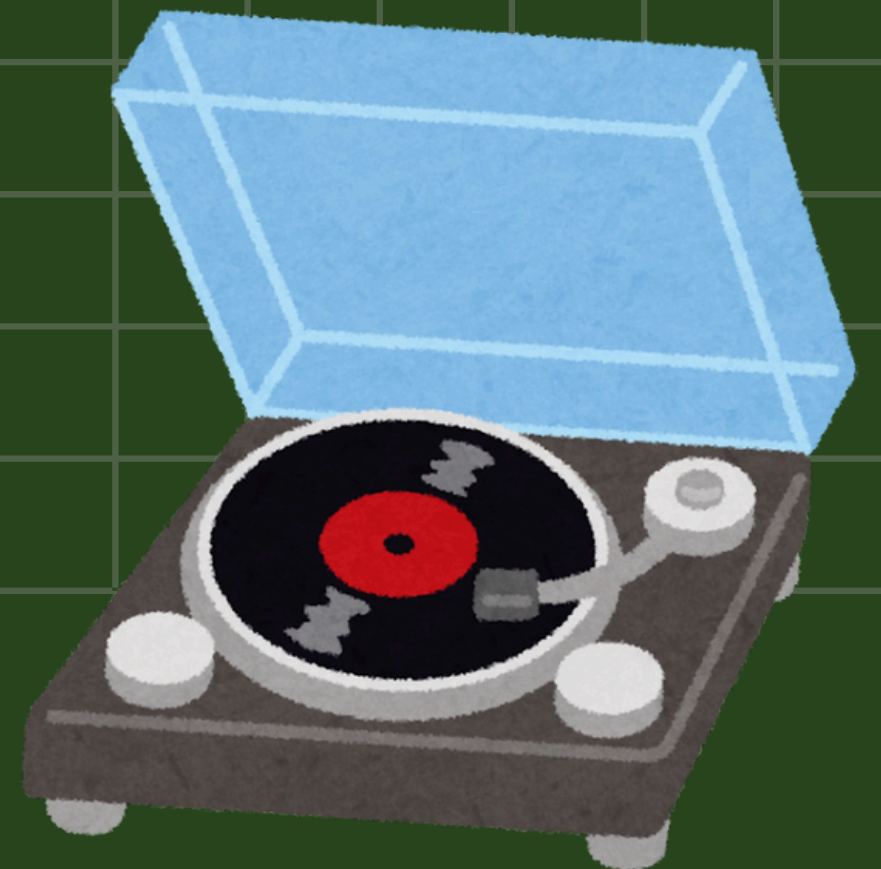
CSCI-635

Group 4

Spotify Tracks: Genre Prediction

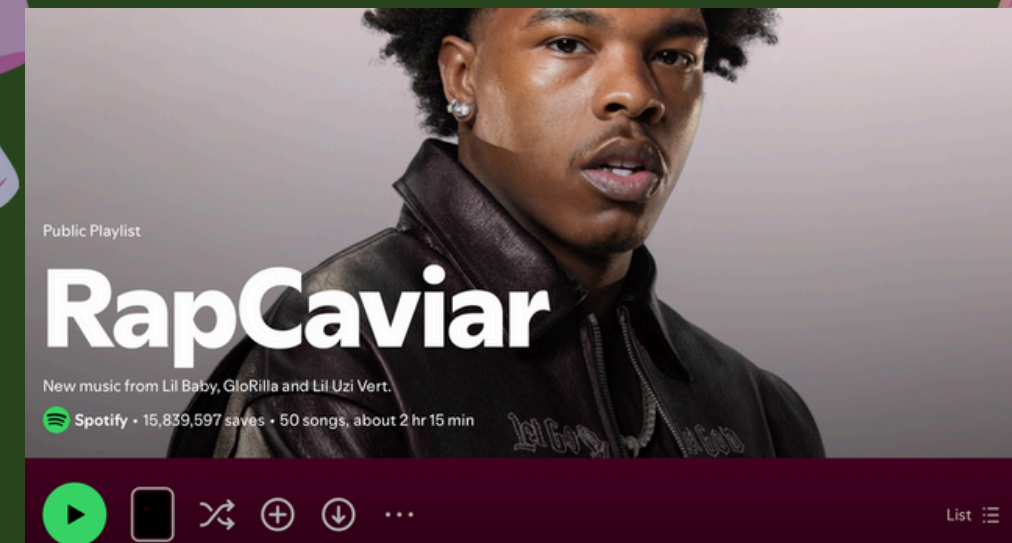
David Jordan & Prionti Nasir

Dec 5th, 2025



Background

- Genre identification allows music streaming platforms to understand what kind of music listeners enjoy listening to
- Aids in the process of building and labeling curated playlists
 - ex) rap caviar by Spotify
- Lots of downstream tasks depend on this
 - targeted marketing, recommendations, artist profiling

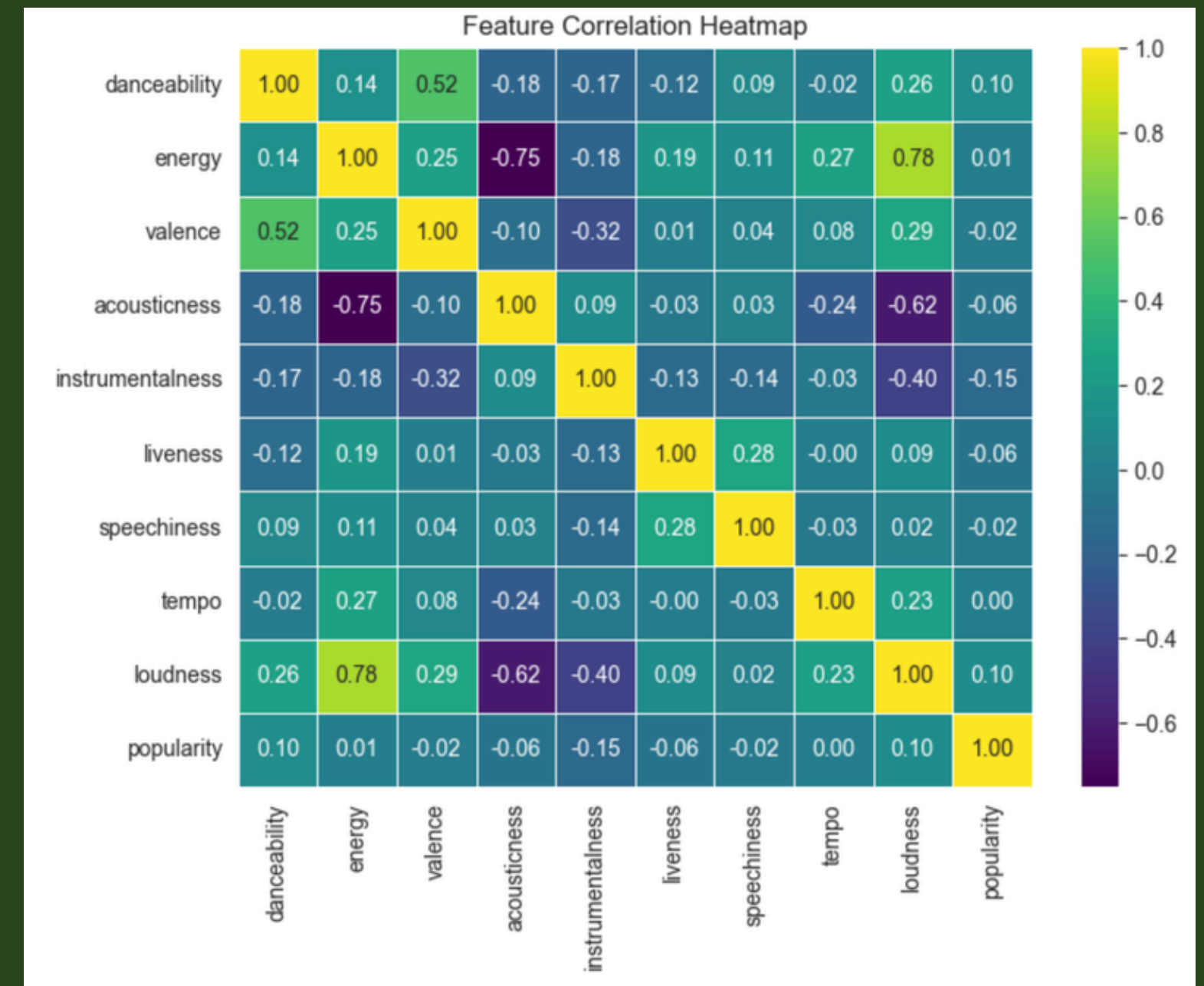


Feature Selection

Spotify 1M Tracks

Acoustic and Rhythmic features

- danceability – how suitable the track is for dancing
- energy – intensity and activity
- valence – musical positivity
- acousticness – acoustic vs electronic
- instrumentalness – absence of vocals
- liveness – probability of live performance
- speechiness – spoken words proportion
- tempo
- loudness
- popularity
- ~~duration~~
- genre – track classification (**target**)

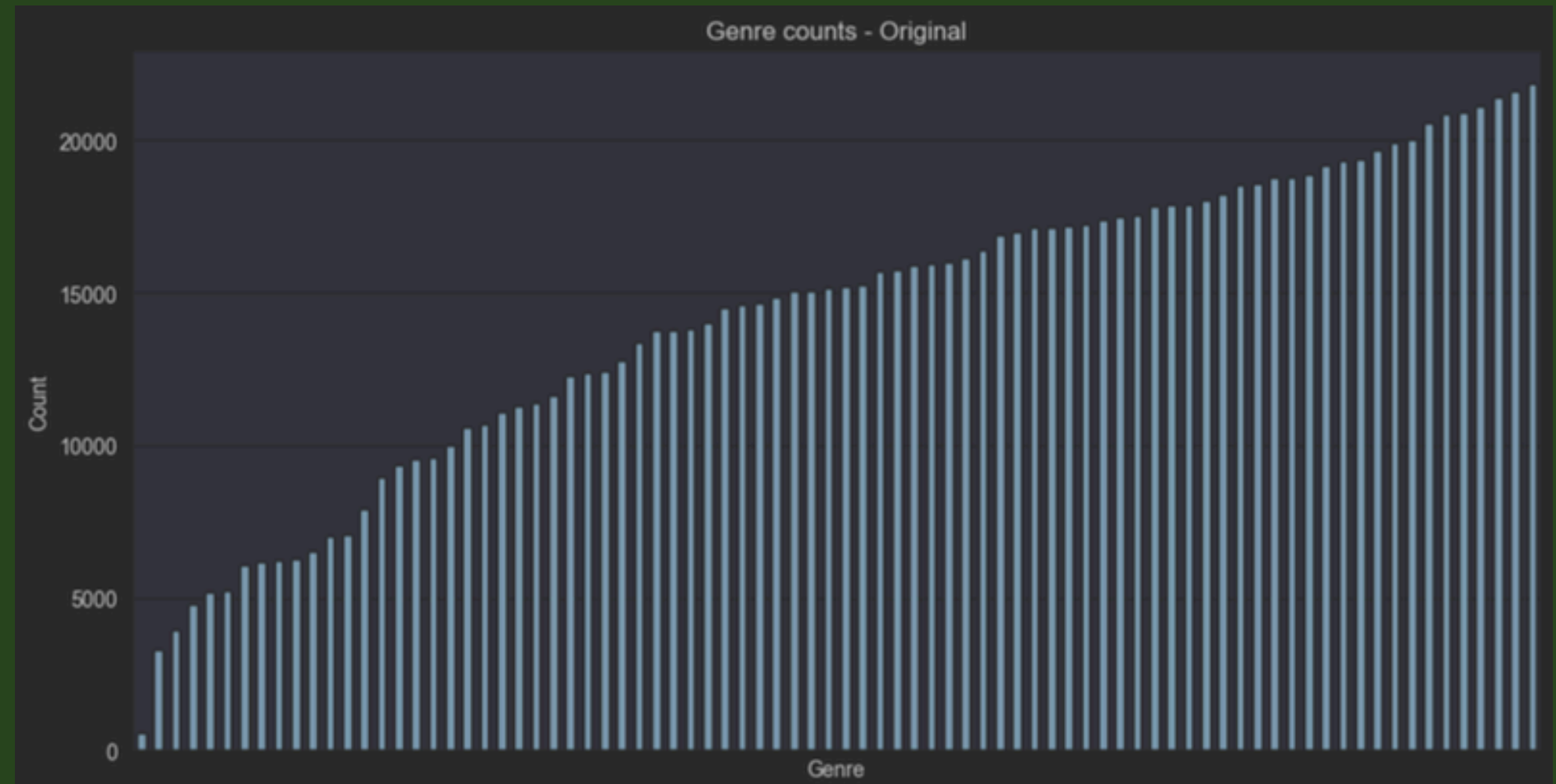


Data Analysis

Spotify 1M Tracks

```
['acoustic' 'afrobeat' 'alt-rock' 'ambient' 'black-metal' 'blues'  
'breakbeat' 'cantopop' 'chicago-house' 'chill' 'classical' 'club'  
'comedy' 'country' 'dance' 'dancehall' 'death-metal' 'deep-house'  
'detroit-techno' 'disco' 'drum-and-bass' 'dub' 'dubstep' 'edm' 'electro'  
'electronic' 'emo' 'folk' 'forro' 'french' 'funk' 'garage' 'german'  
'gospel' 'goth' 'grindcore' 'groove' 'guitar' 'hard-rock' 'hardcore'  
'hardstyle' 'heavy-metal' 'hip-hop' 'house' 'indian' 'indie-pop'  
'industrial' 'jazz' 'k-pop' 'metal' 'metalcore' 'minimal-techno'  
'new-age' 'opera' 'party' 'piano' 'pop' 'pop-film' 'power-pop'  
'progressive-house' 'psych-rock' 'punk' 'punk-rock' 'rock' 'rock-n-roll'  
'romance' 'sad' 'salsa' 'samba' 'sertanejo' 'show-tunes'  
'singer-songwriter' 'ska' 'sleep' 'songwriter' 'soul' 'spanish' 'swedish'  
'tango' 'techno' 'trance' 'trip-hop']
```

82 genres (classes)



songwriter (589)
black-metal (21,837)

Initial Approach

- It is unrealistic to expect any meaningful results attempting to classify 82 genres
- → we attempted using the top 20 genres, resulting in an average **accuracy of 47%** across our Decision Tree and SVM models

```
['acoustic' 'afrobeat' 'alt-rock' 'ambient' 'black-metal' 'blues'  
'breakbeat' 'cantopop' 'chicago-house' 'chill' 'classical' 'club'  
'comedy' 'country' 'dance' 'dancehall' 'death-metal' 'deep-house'  
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'tango' 'techno' 'trance' 'trip-hop']
```

Clustering to Merge Subgenres

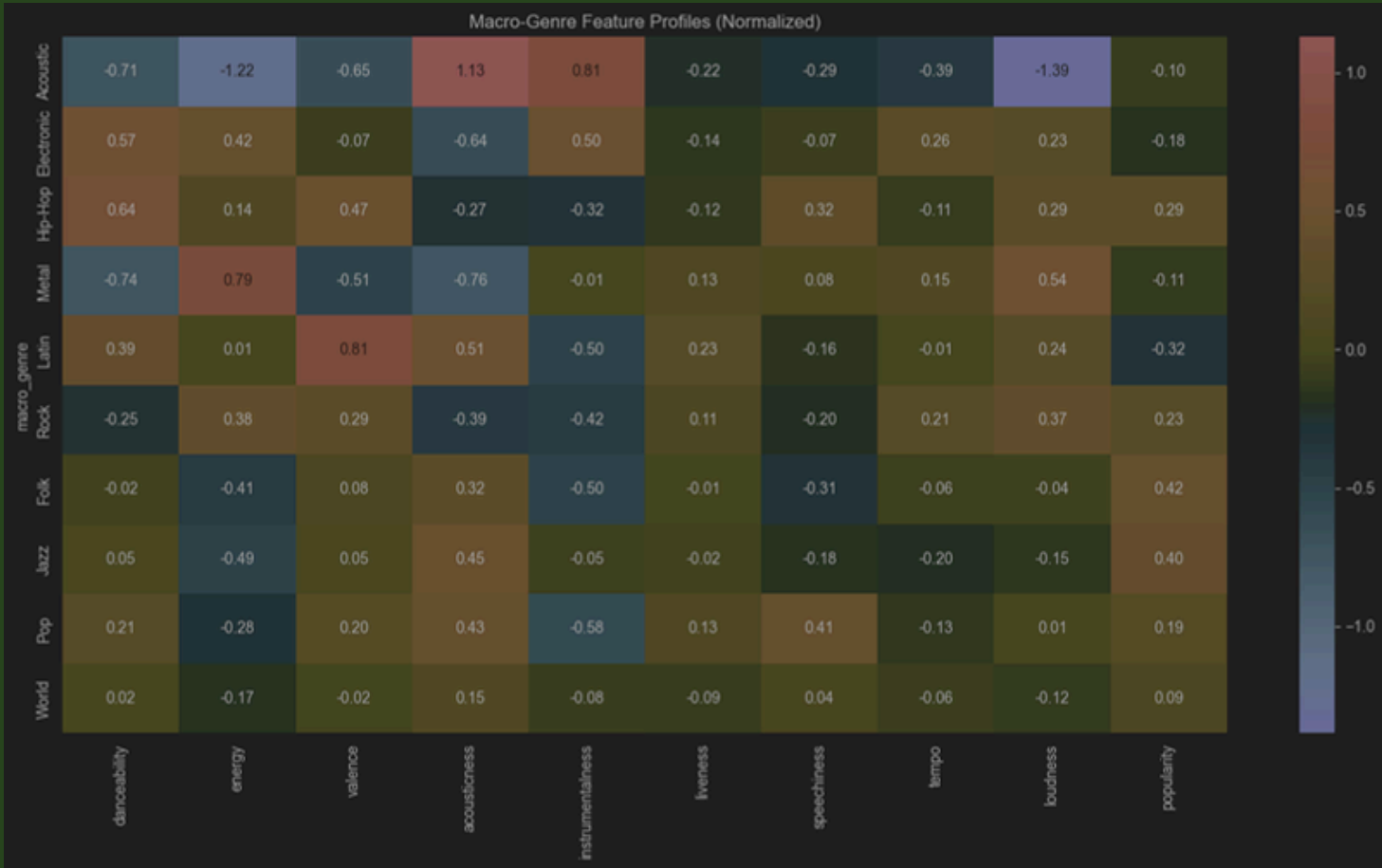
We use hierarchical clustering to identify acoustically similar genres.
Genres that merge at low distances are acoustically similar.

Clustering to merge subgenres

Auto Cluster	Contents	Problem
Cluster 9	17 genres: gospel, german, swedish, spanish, soul, folk, blues, country, k-pop, chill, french, funk, electro, sad, indie-pop, jazz, electronic	Algorithm couldn't separate these
Cluster 7	pop, hip-hop, rock, dance	4 completely different genres lumped together

Groupings: acoustically correct but contextually nonsensical

- Cluster 2: guitar, ambient, sleep, new-age, piano, classical (acoustic/calm) ✓
- Cluster 3: dub, trip-hop, minimal-techno, house, trance, dubstep, edm, detroit-techno, groove, progressive-house, club, chicago-house, breakbeat, deep-house, techno (electronic) ✓
- Cluster 9: gospel, german, swedish, spanish, soul, folk, blues, country, k-pop, chill, french, funk, electro, sad, indie-pop, jazz, electronic ✗



Final Selection for Classification

Rock

rock
alt-rock
hard-rock
rock-n-roll
punk
punk-rock
psych-rock
power-pop
ska

Hip-Hip

funk
groove
afrobeat
soul
dancehall

Jazz

Jazz
Gospel

World

Indian
French
German
Swedish

Pop/Other

Pop
Pop-film
party
romance
sad
show-tunes
comedy
k-po
indie-pop
cantopop

Folk/Country

folk
country
blues
singer-
songwriter

Acoustic

classical
piano
guitar
opera
Ambient
Acoustic
Chill
Sleep
new-age

Latin

Salsa
Samba
Sertanejo
Forro
Tango
Spanish

Electronic

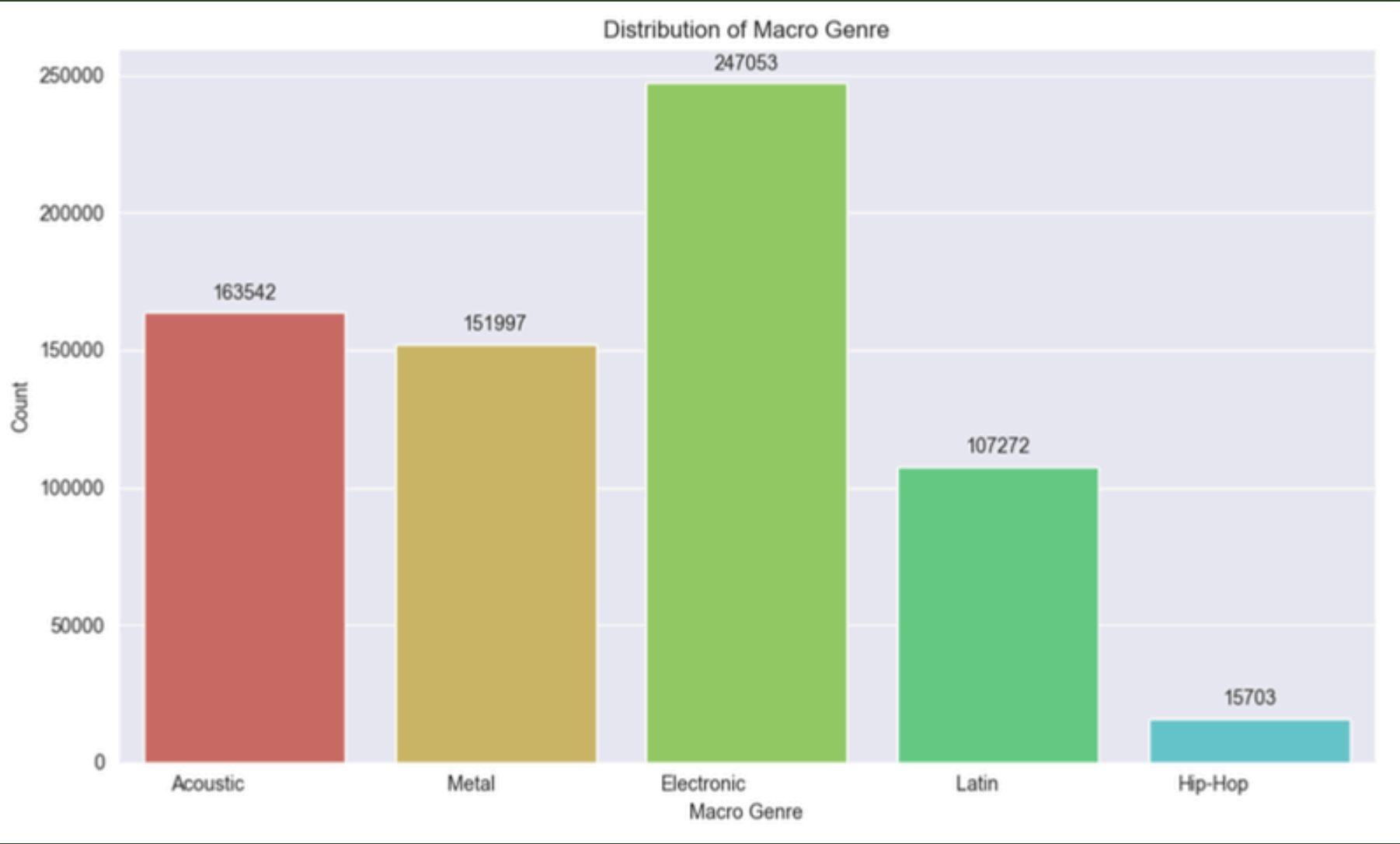
Deep-house	techno
Minimal-techno	house
drum-and-bass	edm
club	trance
dance	dubstep
disco	progressive-house
electronic	dub
chicago-house	
detroit-techno	
electro	
hardstyle	
breakbeat	
garage	
trip-hop	

Metal

Metal	
black-metal	grindcore
death-metal	hardcore
heavy-metal	industrial
metalcore	goth
	emo

Hip-Hop

Distribution



Acoustic

- classical
- piano
- guitar
- opera
- Ambient
- Acoustic
- Chill
- Sleep
- new-age

Electronic

- Deep-house
- Minimal-techno
- drum-and-bass
- club
- dance
- disco
- electronic
- techno
- house
- edm
- trance
- dubstep
- progressive-house
- dub
- chicago-house
- detroit-techno
- electro
- hardstyle
- breakbeat
- garage
- trip-hop

Latin

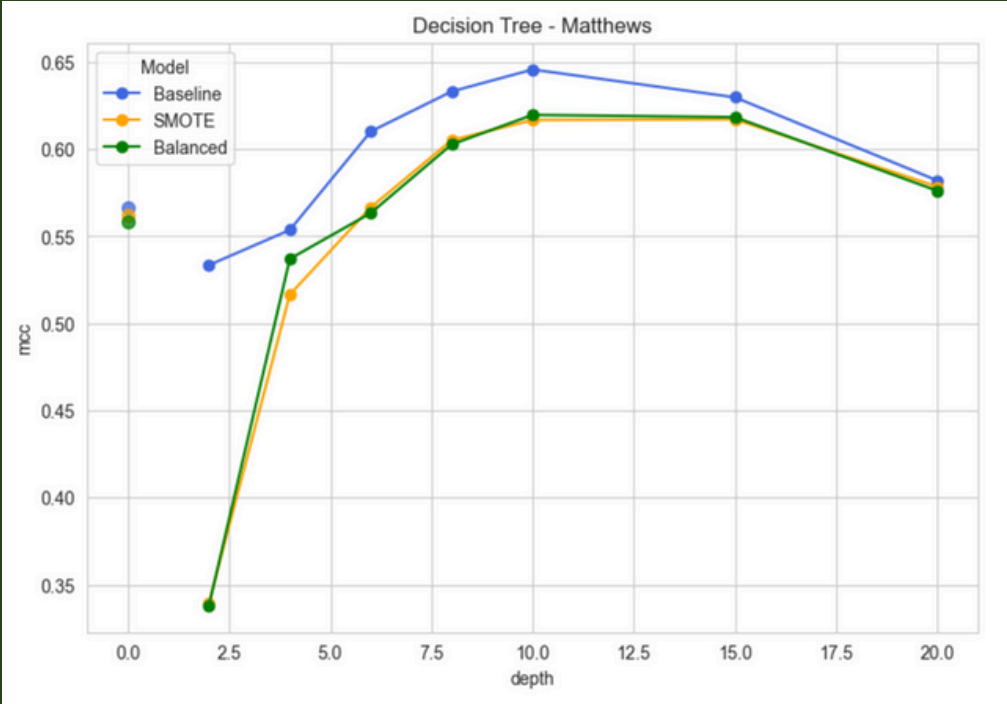
- Salsa
- Samba
- Sertanejo
- Forro
- Tango
- Spanish

Metal

- Metal
- black-metal
- death-metal
- heavy-metal
- metalcore
- grindcore
- hardcore
- industrial
- goth
- emo

Hip-Hop

Decision Tree and Gradient Boosted Tree



Smote Report:

	precision	recall	f1-score	support
0	0.78	0.73	0.76	40886
1	0.81	0.63	0.71	61763
2	0.29	0.79	0.43	3926
3	0.64	0.77	0.70	26818
4	0.69	0.76	0.72	37999

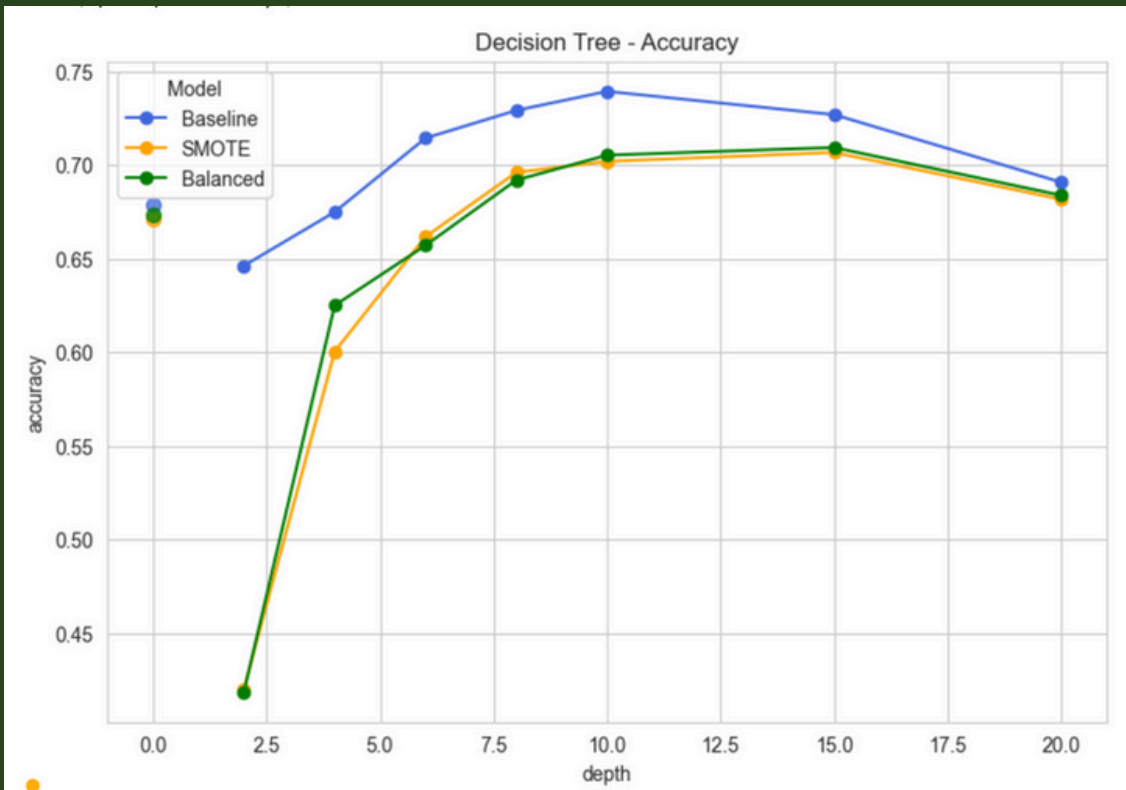
accuracy 0.71 171392

Gradient Boosted Tree

Baseline: 75.69%

Smote: 73.24%

Balanced class weights: 73.20%



Balanced class weight:

	precision	recall	f1-score	support
0	0.81	0.74	0.77	40886
1	0.81	0.62	0.71	61763
2	0.24	0.89	0.38	3926
3	0.64	0.76	0.69	26818
4	0.70	0.75	0.72	37999

accuracy 0.70 171392

Baseline Report:

	precision	recall	f1-score	support
0	0.80	0.75	0.77	40886
1	0.73	0.78	0.75	61763
2	0.52	0.47	0.50	3926
3	0.69	0.73	0.71	26818
4	0.75	0.70	0.73	37999

accuracy

0.74 171392

Logistic Regression

Baseline Logistic Regression Results

Accuracy: 70.04%

F1 Score (macro): 0.5824

Classification Report:

precision recall f1-score support

Acoustic	0.80	0.73	0.76	32730
Electronic	0.66	0.77	0.71	47171
Hip-Hop	0.31	0.04	0.07	5288
Latin	0.68	0.67	0.68	21562
Metal	0.69	0.70	0.69	30363

accuracy				0.70	137114
macro avg	0.63	0.58	0.58	137114	
weighted avg	0.69	0.70	0.69	137114	

Class-Balanced Logistic Regression Results

Accuracy: 65.73%

F1 Score (macro): 0.6065

Classification Report:

precision recall f1-score support

Acoustic	0.82	0.70	0.75	32730
Electronic	0.78	0.53	0.63	47171
Hip-Hop	0.17	0.51	0.26	5288
Latin	0.61	0.75	0.67	21562
Metal	0.66	0.77	0.71	30363

accuracy				0.66	137114
macro avg	0.61	0.65	0.61	137114	
weighted avg	0.71	0.66	0.67	137114	

SMOTE Logistic Regression Results

Accuracy: 65.54%

F1 Score (macro): 0.6052

Classification Report:

precision recall f1-score support

Acoustic	0.82	0.70	0.75	32730
Electronic	0.78	0.53	0.63	47171
Hip-Hop	0.17	0.52	0.26	5288
Latin	0.61	0.75	0.67	21562
Metal	0.66	0.77	0.71	30363

accuracy				0.66	137114
macro avg	0.61	0.65	0.61	137114	
weighted avg	0.71	0.66	0.67	137114	



MLP

Baseline MLP Results

Accuracy: 74.50%

F1 Score (macro): 0.6616

Classification Report:

precision recall f1-score support

Acoustic	0.80	0.77	0.78	32730
Electronic	0.72	0.81	0.76	47171
Hip-Hop	0.45	0.21	0.29	5288
Latin	0.74	0.71	0.73	21562
Metal	0.76	0.74	0.75	30363

Class-Balanced MLP Results

Accuracy: 74.50%

F1 Score (macro): 0.6616

Classification Report:

precision recall f1-score support

Acoustic	0.80	0.77	0.78	32730
Electronic	0.72	0.81	0.76	47171
Hip-Hop	0.45	0.21	0.29	5288
Latin	0.74	0.71	0.73	21562
Metal	0.76	0.74	0.75	30363

SMOTE MLP Results

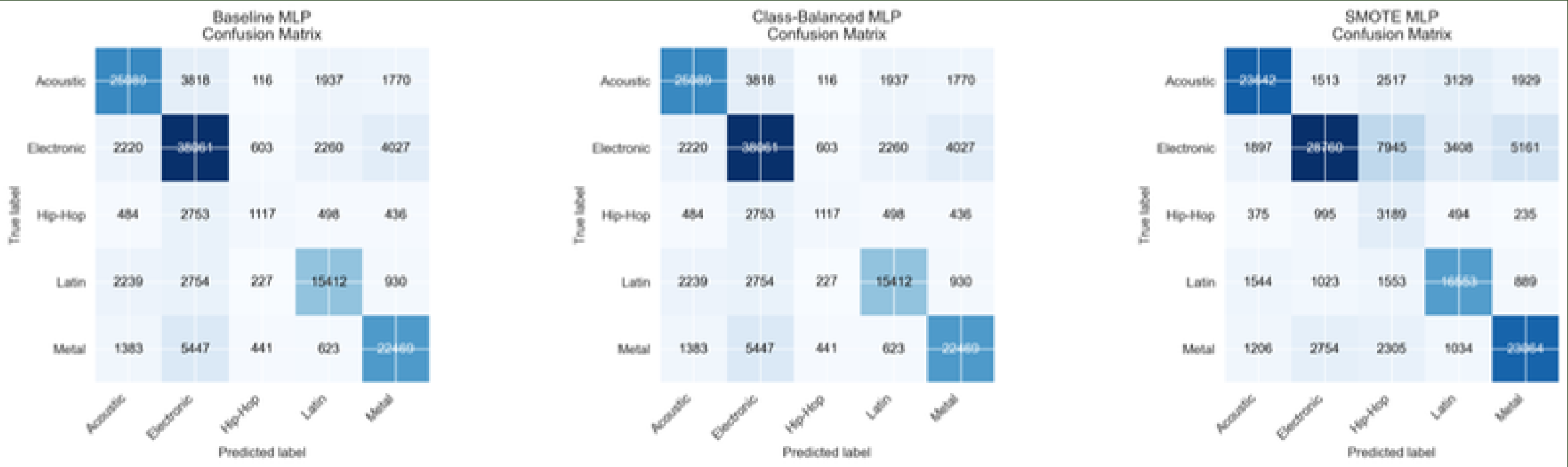
Accuracy: 69.44%

F1 Score (macro): 0.6430

Classification Report:

precision recall f1-score support

Acoustic	0.82	0.72	0.77	32730
Electronic	0.82	0.61	0.70	47171
Hip-Hop	0.18	0.60	0.28	5288
Latin	0.67	0.77	0.72	21562
Metal	0.74	0.76	0.75	30363



Improvements



Song

Build It Better



Aron Wright • Build It Better • 2015 • 3:54 • 23,731,151

- **Top-k accuracy**
- **Better class reduction/sampling techniques**
- **Deeper NN architectures:** a successful project used 256→128→64→32 structure with dropout and batch normalization
- **Ensemble methods**
- **Additional features:** Incorporating lyrics-based or spectral features beyond Spotify's audio metrics

Thank You

Gradient boosted trees explained: https://www.youtube.com/watch?v=hjxgoUJ_va8

Gradient boosted trees in python: <https://www.youtube.com/watch?v=ImghItTFelw>

Spotify 1m Tracks dataset: <https://www.kaggle.com/datasets/amitanshjoshi/spotify-1million-tracks>

Genre prediction using audio features: <https://medium.com/@navodas88/music-genre-prediction-using-audio-features-96dea4e71ad3>

Machine Learning Approach for Genre Prediction on Spotify: <https://cdr.lib.unc.edu/downloads/8k71nn03b?locale=en>

Solving Spotify Multiclass Genre Classification Problem: <https://www.analyticsvidhya.com/blog/2023/03/solving-spotify-multiclass-genre-classification-problem/>